

Questions on Exception Handling

1. Write a program to handle a `ZeroDivisionError` when dividing a number by zero.
2. Prompt the user for an integer and raise `ValueError` if input is invalid.
3. Open a non-existent file and handle `FileNotFoundError`.
4. Take two user inputs, validate they are numbers, else raise `TypeError`.
5. Open a restricted file and handle `PermissionError`.
6. Access an invalid list index and handle `IndexError`.
7. Prompt input and handle `KeyboardInterrupt` if user cancels it (Ctrl+C).
8. Perform division and handle `ArithmeticError`.
9. Read a file with wrong encoding and handle `UnicodeDecodeError`.
10. Use a string method on a list and handle `AttributeError`.
11. Try to open a dictionary key that doesn't exist and handle `KeyError`.
12. Access a non-existent value in a set and handle the exception.
13. Try converting a string to float and handle `ValueError`.
14. Try importing a non-existent module and handle `ModuleNotFoundError`.
15. Open a file for writing in a read-only directory and handle `IOError`.
16. Create a program that catches multiple exceptions using a single try block.
17. Use nested try-except blocks to handle inner and outer-level exceptions.
18. Demonstrate finally block always running, even after exception.
19. Use else with try-except and explain its behavior with a small code.
20. Simulate a network request and raise a `TimeoutError` (use custom exception).
21. Define your own custom exception class `NegativeNumberError` and raise it.
22. Use a try-except block inside a loop that processes multiple inputs.
23. Raise an exception manually using `raise` when age is below 18.
24. Simulate a simple ATM: Handle wrong PIN (`ValueError`), insufficient balance (`RuntimeError`), and invalid menu input.
25. Write a function that calculates square root and handles negative inputs (`ValueError`).
26. Handle `TypeError` when adding incompatible data types (e.g., string + int).

27. Open a CSV file and handle both `FileNotFoundError` and `UnicodeDecodeError`.
28. Simulate a database connection failure using custom exception `DatabaseConnectionError`.
29. Use exception chaining to raise a new exception while preserving the original.
30. Simulate a file copy operation and handle `IOError`, `FileNotFoundError`, and `PermissionError` together.